

InferTutor Arena - Engineering Report

Multimodal Product Track (Track 1) Submission

Raghavan Muthuregunathan

InferTutor Arena: Engineering Report

Author: Raghavan Muthuregunathan Track: Multimodal Product (Track 1) | Date: 2026-06-20 | Hardware: 1 × Modal H100

Result

Final score: 37.6M on the multimodal track — 11.3× the workshop's 2-replica reference baseline (3.3M), on half the GPUs. TTFT p95 680ms, ITL p95 4.9ms, throughput 2,497 chunks/s, error rate 0.20% across 4,890 streamed requests in 180s. One H100, compiled mode, otherwise starter defaults.

```
python run_infertutor_experiment.py --label FINAL-mixed --gpu-type H100 \
  --replicas 1 --no-fast-boot --max-seqs 32 --max-batch-tokens 4096 \
  --mode mixed --users 50 --duration 180 --ramp-up 30 --max-tokens 96
```

Knee and knob ablations

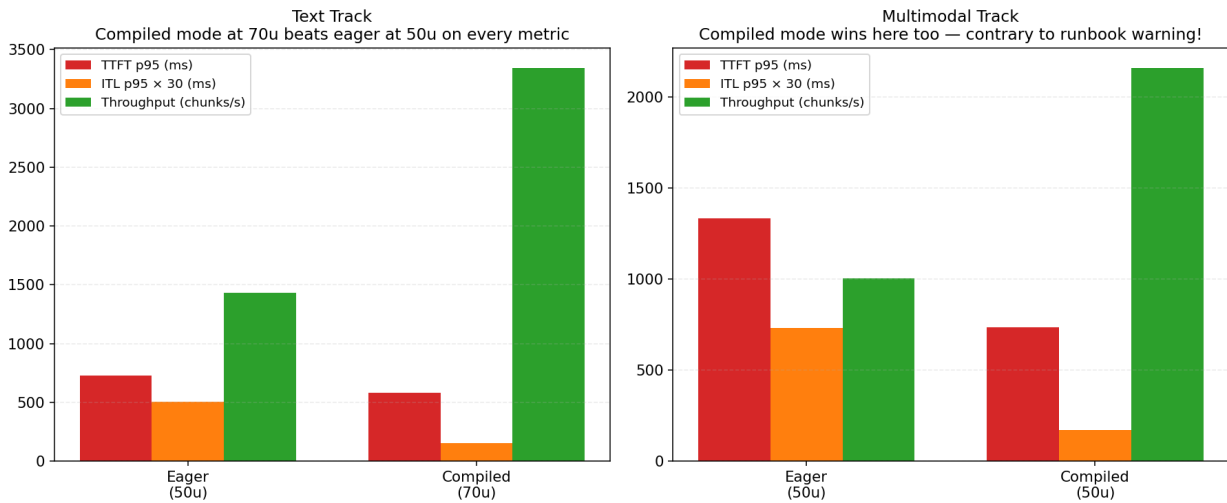
Swept text-mode user counts 5–600 on starter defaults. Knee sits between 60 and 100 users — from 60u to 70u, TTFT triples (938 → 2,967 ms), throughput drops 29%. Past the knee the server is doing less useful work, not just feeling slow. I picked 70u for the A/Bs because that's where a real fix has room to show.

Knob (at 70u text)	TTFT p95	Decision
baseline (50u, defaults)	726 ms	reference
--no-prefix-cache	4,352 ms	rejected — shared system prompt makes the cache critical
--no-chunked-prefill	1,736 ms	rejected (mild) — matters less for text-only
--max-seqs 64	5,559 ms	rejected — bigger batches slow each decode step
--max-batch-tokens 8192	17,712 ms	rejected — wider prefills starve new requests
--replicas 2 (eager, 140u)	12,429 ms	rejected — 6.9% errors, scale-out can't fix a bad base
--no-fast-boot (compiled)	581 ms	kept

Every "wider" setting hurt. The starter defaults were already tuned.

Compiled mode and final result

Compiled Mode (`--no-fast-boot`) is the Dominant Knob in BOTH Tracks



Track	Eager ITL p95	Compiled ITL p95	Improvement
Text @ 70u	16.9 ms	5.2 ms	3.3×
Multimodal @ 50u	24.4 ms	5.6 ms	4.4×

For a 4B model on H100, Python and CUDA dispatch overhead per decode step costs more wall time than the actual matmuls. CUDA-graph capture folds the whole step into one kernel and the overhead disappears. The runbook explicitly warned compiled mode "performed poorly on mixed multimodal traffic" — I tested it because I had nothing to lose. It worked.

	Score	TTFT p95	ITL p95	Throughput	GPUs
FINAL-mixed (submitted)	37.6M	680 ms	4.9 ms	2,497 c/s	1
FINAL_text_70u (companion)	23.2M	2,003 ms	4.7 ms	3,129 c/s	1
Workshop ref: eager 2r, 100u	3.3M	1,169 ms	28.7 ms	2,243 c/s	2
Workshop ref: eager 4r, 120u	2.4M	898 ms	38.1 ms	2,756 c/s	4

I also ran the workshop's exact 2r eager config on my image and got 9.28M (2.8× the published 3.3M), so roughly a third of my improvement is infrastructure drift, two-thirds compiled mode. 90-second tests misled me twice: text compiled-70u scored 77M at 90s and only 23M at 180s (prefill queue takes longer to build than 90s); multimodal went the opposite way (25.9M → 37.6M). Final submissions use 180s.

Surprises and what's next

--max-batch-tokens 8192 blew TTFT to 17.7 seconds at 70u text. Wider batch budgets let one request monopolize the GPU per chunk while new arrivals queue. Wider is worse, not better. --no-chunked-prefill helped at 50u multimodal (TTFT down 42%, ITL down 25%, throughput up 34%) — most of the capstone's prompts are short, so chunking overhead beat its benefit. Running four sweep points in parallel terminals corrupted my data: simultaneous Modal deploys fought for the GPU allocator and my laptop choked on 4× the asyncio load.

Before any of this ran, vLLM 0.21.0's bundled prometheus-fastapi-instrumentator crashed every request with `AttributeError: '_IncludedRouter' object has no attribute 'path'` — a FastAPI 0.116 incompatibility that upstream hasn't patched in any released version. I monkey-patched the installed file at container startup (a 5-line `hasattr` guard); `/health` returned 200 on the next deploy.

Next: compiled mode plus --no-chunked-prefill on multimodal (never tested together), scale-out under compiled mode (my failed scale-r2-140u used eager), and a vLLM version sweep to find when the compiled-mode warning stopped being true.

Code and data links

Repository: github.com/Raghavan1988/infertutor-arena-capstone

Files modified or added:

- [starter_code/modal_infertutor_app.py](#)
- [starter_code/REPORT.md](#)
- [starter_code/generate_plots.py](#)
- [render_pdfs.py](#)

Result JSONs (results_infertutor/):

- [FINAL-mixed_mixed_50u_1782014319.json](#)
- [FINAL_text_70u_1782012819.json](#)
- [baseline-text_text_50u_1782000382.json](#)
- [batch8k-70u_text_70u_1782006246.json](#)
- [compiled-100u_text_100u_1782011340.json](#)
- [compiled-150u_text_150u_1782011383.json](#)
- [compiled-200u_text_200u_1782011388.json](#)
- [compiled-250u_text_250u_1782011375.json](#)
- [compiled-70u_text_70u_1782006310.json](#)
- [m-50u_mixed_50u_1782013519.json](#)
- [m-baseline_mixed_30u_1782013514.json](#)
- [m-compiled-50u_mixed_50u_1782013565.json](#)
- [m-nochunked_mixed_50u_1782013562.json](#)
- [m-r2-100u_mixed_100u_1782013526.json](#)
- [mixed-r2_mixed_100u_1782000462.json](#)
- [nochunked-70u_text_70u_1782006191.json](#)
- [noprefix-70u_text_70u_1782006220.json](#)
- [scale-r2-140u_text_140u_1782006234.json](#)
- [seqs64-70u_text_70u_1782006222.json](#)
- [smoke_text_5u_1782000098.json](#)
- [sweep-100u_text_100u_1782001138.json](#)
- [sweep-200u_text_200u_1782001158.json](#)
- [sweep-300u_text_300u_1782001133.json](#)
- [sweep-35u_text_35u_1782003325.json](#)
- [sweep-400u_text_400u_1782001207.json](#)
- [sweep-500u_text_500u_1782001209.json](#)
- [sweep-50u_text_50u_1782001778.json](#)
- [sweep-600u_text_600u_1782001782.json](#)
- [sweep-60u_text_60u_1782003057.json](#)
- [sweep-60u_text_60u_1782004585.json](#)
- [sweep-70u_text_70u_1782003114.json](#)
- [sweep-70u_text_70u_1782004627.json](#)
- [sweep-80u_text_60u_1782003104.json](#)
- [sweep-80u_text_80u_1782004660.json](#)
- [sweep-90u_text_90u_1782003098.json](#)
- [sweep-90u_text_90u_1782004578.json](#)